

AREHI-SECOPS

ASEAN Regional E-Commerce Hybrid Infrastructure & Hardened SecOps System

Phase 2 — Dynamic Routing & IPv6

Full-Depth Configuration & Certification Study Guide

Architect / Author: Danial Nur Irfan bin Azeze

Bachelor of Computer Science — Networks & Security, UTM

CompTIA Security+ Certified

Exam objectives referenced: Cisco CCNA 200-301, CompTIA Security+ SY0-701, AWS Certified Solutions

Architect – Associate (SAA-C03)

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Devices in scope (the 4 that route):

MY-KL-HQ-CORE — Catalyst 3650-24PS

SG-EDGE-GW — ISR 4331

PH-MNL-ROAS / TH-BKK-ROAS — ISR 4321

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1. Phase 2 Overview & Why It Exists

Phase 1 ended with every site able to reach Malaysia HQ directly, but no site able to reach any *other* site — Manila couldn't talk to Bangkok, for example — because reachability depended entirely on static default routes pointing only toward HQ. Phase 2 replaces that stopgap with genuine dynamic routing (OSPF Area 0), and adds IPv6 dual-stack addressing across the same four routing devices. By the end of this phase, every site can reach every other site's internal subnets, learned automatically, with zero static routes holding it together.

Devices in scope

Hostname	Role	Confirmed PT model
MY-KL-HQ-CORE	Core/distribution switch (Layer 3) — the OSPF hub	Catalyst 3650-24PS
SG-EDGE-GW	WAN edge router — single-homed OSPF spoke	ISR 4331
PH-MNL-ROAS	Router-on-a-stick — Manila's OSPF spoke	ISR 4321
TH-BKK-ROAS	Router-on-a-stick — Bangkok's OSPF spoke	ISR 4321

PH-MNL-ACC and TH-BKK-ACC (the two Layer 2 access switches) are absent from this entire phase — neither runs OSPF nor gets IPv6 addressing, since they're pure Layer 2 devices that simply rely on their local ROAS router's default gateway for anything off-subnet.

2. A Note on Exam Objective Tags

Phase 2 is almost entirely **CCNA 200-301 Domain 3.0 (IP Connectivity)** territory — OSPF, FHRP-adjacent concepts, and IPv6 addressing are core CCNA subject matter. Unlike Phase 1 and Phase 3, there is genuinely **no direct Security+ objective** exercised in this phase — no authentication, encryption, or hardening control is configured here (OSPF authentication, for instance, is not part of this phase's scope). There is also **no direct AWS SAA-C03 tie** — dynamic routing and IPv6 addressing on-premises don't map onto any SAA-C03 exam objective the way Phase 1's cloud-facing placeholder interface did. Rather than force tags onto every objective for the sake of appearing balanced, this guide tags Phase 2 honestly: it's a CCNA-focused phase.

3. Objective 1 — IPv6 Addressing Plan & OSPF Router ID/Loopback Plan

Planning only — no device commands, this objective produced the reference plan the rest of Phase 2 is built from

CCNA 200-301: Domain 1.0 (Network Fundamentals — IPv6 addressing) & Domain 3.0 (IP Connectivity)

Purpose: before typing a single OSPF or IPv6 command, decide the addressing scheme and Router ID assignment up front — exactly the same reasoning as building the IPv4 VLSM plan before Phase 1's device configuration began. Skipping this step and improvising addresses per-device risks overlapping subnets or inconsistent Router IDs discovered only after something breaks.

IPv6 addressing plan

Parent block: `2001:db8::/32` — defined by **RFC 3849** as a documentation-only prefix, the IPv6 equivalent of the IPv4 `203.0.113.0/24` placeholder used in Phase 1's cloud-facing interface. Each site gets a `/48` ; each VLAN within a site gets a `/64` carved out of it.

Term	Meaning
<code>/48</code>	The standard allocation size for an entire site/organization in IPv6 addressing — leaves 16 bits of subnetting room underneath for individual VLANs, far more headroom than IPv4 ever provides at a comparable tier.
<code>/64</code>	Not an arbitrary choice — <code>/64</code> is mandatory for any subnet that needs to use SLAAC (Stateless Address Autoconfiguration) or modified EUI-64, both of which require exactly 64 bits of interface identifier space. There is no finer VLSM subdivision possible at the LAN edge in IPv6 the way there is in IPv4.
<code>/127</code>	Defined by RFC 6164 as modern best practice for point-to-point WAN links — the IPv6 equivalent of using a IPv4 <code>/30</code> instead of wastefully handing out a full <code>/64</code> to a link that only ever has 2 devices on it.

Site	IPv6 site block
Malaysia (HQ)	<code>2001:db8:1::/48</code>
Singapore (WAN edge, no local VLANs)	<code>2001:db8:2::/48</code>
Manila	<code>2001:db8:3::/48</code>
Bangkok	<code>2001:db8:4::/48</code>
WAN transit links (cross-site, belongs to no single site)	<code>2001:db8:ff::/48</code>

Site	VLAN	IPv6 subnet
Malaysia	10 / 20 / 30 / 40	2001:db8:1:10::/64 / :20::/64 / :30::/64 / :40::/64
Manila	10 / 20 / 30 / 40	2001:db8:3:10::/64 / :20::/64 / :30::/64 / :40::/64
Bangkok	10 / 20 / 30 / 40	2001:db8:4:10::/64 / :20::/64 / :30::/64 / :40::/64

WAN link	IPv6 subnet	MY side	Remote side
MY ↔ Singapore	2001:db8:ff::0/127	::0	::1
MY ↔ Manila	2001:db8:ff::2/127	::2	::3
MY ↔ Bangkok	2001:db8:ff::4/127	::4	::5

EXAM ANGLE — LINK-LOCAL VS GLOBAL UNICAST

Every routed interface additionally gets its automatic **link-local** address (fe80:::/10 , generated via modified EUI-64 from the interface MAC) regardless of the global unicast address assigned to it. This matters because OSPFv3's neighbor relationships actually form over link-local addresses, not the global address — a frequently tested distinction.

OSPF Router ID & Loopback plan

Router IDs are set explicitly rather than left to auto-election. Each OSPF-speaking device gets a dedicated Loopback0 purely for this purpose.

Term	Meaning
Router ID	A 32-bit value, formatted like an IPv4 address, that uniquely identifies a router within the OSPF domain — used to break ties in DR/BDR elections and to track which router originated which piece of topology information.
Loopback interface	A virtual interface that exists purely in software — it can never physically go down the way a real interface can (a cable can't be unplugged from something that isn't physical), which is exactly why it's the right place to anchor something that needs to stay stable, like a Router ID or, later, an SSH management source address.

Device	Loopback0 (Router ID)
MY-KL-HQ-CORE	10.255.255.1/32
SG-EDGE-GW	10.255.255.2/32
PH-MNL-ROAS	10.255.255.3/32
TH-BKK-ROAS	10.255.255.4/32

DELIBERATELY OUT OF SCOPE

The original project blueprint mentions a floating static route for backup-ISP failover. This topology has no second internet-facing link anywhere — SG-EDGE-GW 's cloud-facing interface is the only WAN-to-cloud path, and it isn't redundant — so there's nothing for a floating static route to fail over to. Deliberately dropped from scope rather than built against a link that doesn't exist.

4. Objective 2 — OSPFv2 Process, Router ID, Network Statements

MY-KL-HQ-CORE, SG-EDGE-GW, PH-MNL-ROAS, TH-BKK-ROAS

CCNA 200-301: Domain 3.0 (IP Connectivity)

Purpose: OSPF (Open Shortest Path First) is a **link-state** interior gateway protocol — every router builds a complete map of the network's topology and independently calculates shortest paths using Dijkstra's algorithm, rather than just trusting a neighbor's hop-count claim the way a distance-vector protocol (like RIP) does. This is what actually replaces Phase 1's static routes with routes that adapt automatically if the topology changes.

Commands, word by word

```
interface Loopback0
  ip address 10.255.255.1 255.255.255.255
!
router ospf 1
  router-id 10.255.255.1
  network 10.255.255.1 0.0.0.0 area 0
  network 10.10.10.0 0.0.0.255 area 0
  network 10.10.254.0 0.0.0.3 area 0
```

Word / term	Meaning
<code>router ospf 1</code>	Starts the OSPF process. The <code>1</code> is a process ID that is only locally significant to this device — it does not need to match between neighboring routers, and a mismatch here is not why two routers would fail to become neighbors. A commonly tested point precisely because it's commonly misunderstood.
<code>router-id 10.255.255.1</code>	Explicitly pins the Router ID rather than relying on automatic selection. Without this command, IOS picks the highest IP on any active loopback first, or failing that the highest active physical interface IP — but only at the moment the OSPF process first starts. Adding a loopback <i>after</i> OSPF is already running does not change an already-selected Router ID until the process restarts — a classic exam trap.
<code>network 10.10.10.0 0.0.0.255 area 0</code>	Despite how it reads, this does not simply "add a subnet to OSPF." It tells OSPF: any active interface whose IP falls within this network+wildcard range should participate in Area 0. The wildcard mask (<code>0.0.0.255</code>) is the inverse of a subnet mask — a <code>0</code> bit means "must match exactly," a <code>1</code> bit means "don't care." A <code>/24</code> subnet mask (<code>255.255.255.0</code>) inverts to wildcard <code>0.0.0.255</code> ; a <code>/30</code> (<code>255.255.255.252</code>) inverts to <code>0.0.0.3</code> ; a <code>/32</code> loopback inverts to <code>0.0.0.0</code> (every bit must match exactly — one specific address, nothing else).
<code>area 0</code>	Area 0 is the mandatory OSPF backbone area — every other area in a multi-area design must connect directly to it. This topology is small enough that everything simply lives in Area 0, a genuine single-area design, which is the simplest and most common real-world case and exactly what CCNA focuses on before touching multi-area concepts.

Full network-statement reference per device

Device	Network statements (all under <code>area 0</code>)
MY-KL-HQ-CORE	10.255.255.1/32 (loopback), 10.10.10.0/24, 10.10.20.0/24, 10.10.30.0/24, 10.10.40.0/24, 10.10.254.0/30, 10.10.254.4/30, 10.10.254.8/30
SG-EDGE-GW	10.255.255.2/32 (loopback), 10.10.254.0/30
PH-MNL-ROAS	10.255.255.3/32 (loopback), 10.10.110.0/24, 10.10.111.0/24, 10.10.112.0/24, 10.10.113.0/24, 10.10.254.4/30
TH-BKK-ROAS	10.255.255.4/32 (loopback), 10.10.120.0/24, 10.10.121.0/24, 10.10.122.0/24, 10.10.123.0/24, 10.10.254.8/30

EXAM ANGLE — WHY LAN SUBNETS ARE INCLUDED EVEN THOUGH HELLOS AREN'T WANTED THERE YET

Every LAN-side subnet is included in the network statements above, not just the WAN links — this is deliberate. Objective 3 (next) layers `passive-interface` on top of this exact configuration to stop OSPF hello packets from going out on those user-facing VLANs, *without* removing them from being advertised. Advertise the subnet, but don't neighbor on it — that split only makes sense once you've seen both halves.

EXPECTED RESULT

```
show ip ospf neighbor  
show ip protocols
```

Neighbor adjacencies should form on the WAN links between `MY-KL-HQ-CORE` and each of the other three, reaching state `FULL`. `show ip protocols` should list the Router ID and every network statement just typed. At this stage, before Objective 3, OSPF would also attempt to form adjacencies on the LAN sides if another OSPF speaker were present there — none is, so this isn't visibly wrong yet, just not yet fully hardened.

5. Objective 3 — Passive-Interface

MY-KL-HQ-CORE, SG-EDGE-GW, PH-MNL-ROAS, TH-BKK-ROAS

CCNA 200-301: Domain 3.0 (IP Connectivity)

Purpose: sending OSPF hello packets out onto an end-user VLAN is wasted bandwidth at best, and a real risk at worst — any device on that VLAN could attempt to speak OSPF and try to form a bogus adjacency. Passive-interface suppresses hellos on an interface while still advertising its subnet — it doesn't remove anything from OSPF's database, it just stops that specific interface from trying to make friends.

Commands, word by word

```
router ospf 1
  passive-interface default
  no passive-interface GigabitEthernet1/0/22
  no passive-interface GigabitEthernet1/0/23
  no passive-interface GigabitEthernet1/0/24
```

Word / term	Meaning
passive-interface default	Marks <i>every</i> interface participating in this OSPF process as passive in one command, rather than listing each one individually — a "deny by default" pattern.
no passive-interface Gi1/0/22	Selectively un-passives just the WAN links that actually need to form neighbors. This ordering (default-deny, then explicitly allow) is fail-safe: any interface added to OSPF later is automatically passive unless someone deliberately excludes it — the safer default for a network that will keep growing.

EXPECTED RESULT

```
show ip protocols
show ip ospf neighbor
```

`show ip protocols` should list a "Passive Interface(s)" section including every LAN-side interface.

`show ip ospf neighbor` should show adjacencies **only** on the WAN links — the same set of neighbors as before Objective 3, since there was nothing on the LAN side to neighbor with anyway. This step's real value is defensive, not something that changes today's neighbor table.

6. Objective 4 — OSPF Cost Tuning & DR/BDR Priority

MY-KL-HQ-CORE, SG-EDGE-GW, PH-MNL-ROAS, TH-BKK-ROAS

CCNA 200-301: Domain 3.0 (IP Connectivity)

Purpose: two related but separate tuning concerns — how OSPF calculates the "cost" of a path, and how it decides which router manages a multi-access segment.

Cost — commands, word by word

```
router ospf 1
  auto-cost reference-bandwidth 10000
```

Word / term	Meaning
OSPF cost	The metric OSPF actually uses to pick a shortest path — by default, calculated as <code>reference bandwidth ÷ interface bandwidth</code> .
<code>auto-cost</code> <code>reference-</code> <code>bandwidth</code>	The default reference bandwidth is a stale 100 Mbps — meaning any interface Gigabit-speed or faster floors to a calculated cost of 1, making every fast link indistinguishable from every other fast link regardless of actual speed. This is a well-known real-world (and exam) gotcha. Raising the reference value to <code>10000</code> (10 Gbps) gives Gigabit links a more meaningful cost of 1000, restoring OSPF's ability to actually distinguish between link speeds.

MUST BE IDENTICAL ACROSS EVERY ROUTER

A mismatched reference bandwidth between two OSPF routers does not break their adjacency — they'll still become neighbors — but it does produce inconsistent path-cost math between them, since each is scaling costs against a different baseline. This single command must be typed identically on all 4 devices.

DR/BDR priority — commands, word by word

```
! MY-KL-HQ-CORE:
interface GigabitEthernet1/0/22
  ip ospf priority 255

! Each spoke (SG-EDGE-GW, PH-MNL-ROAS, TH-BKK-ROAS):
ip ospf priority 0
```

Word / term	Meaning
DR / BDR (Designated Router / Backup Designated Router)	On a multi-access broadcast network segment, OSPF elects one router as DR to reduce the number of adjacencies every router would otherwise need to maintain with every other router on that segment. Critically, OSPF treats Ethernet as a broadcast network type by default regardless of how many routers are actually on the segment — even a link with only 2 routers still runs a full DR/BDR election, which is a genuine exam trap.
<code>ip ospf priority 255</code>	The maximum possible priority — guarantees this router wins the DR election on this interface. Set on <code>MY-KL-HQ-CORE</code> since it's the hub every spoke only ever talks to.
<code>ip ospf priority 0</code>	The special value meaning "never a DR/BDR candidate at all," regardless of what else is on the segment. Set on every spoke, since a spoke router has no other neighbors on that link to serve as DR for anyway.

A SUBTLETY WORTH KNOWING: NO BDR AT ALL

With only 2 routers on each WAN link and the spoke's priority set to 0, the segment ends up with a DR (`MY-KL-HQ-CORE`) but genuinely **no BDR** — the only other router present refuses to be a candidate. This is a valid, intentional outcome here: if the hub goes down, the entire link is down regardless of DR status, so a backup DR on a 2-router point-to-point-in-practice link provides no real value.

EXPECTED RESULT

```
show ip ospf interface GigabitEthernet1/0/22
```

Confirms the recalculated cost reflects the new reference bandwidth, and that the priority value and DR/BDR role shown match what was configured — `MY-KL-HQ-CORE` should report itself as `DR` on each WAN link; every spoke should report `DROTHER` , never DR or BDR.

7. Objective 5 — Retiring Static Routes & Proving Cross-Site Reachability

PH-MNL-ROAS, TH-BKK-ROAS for the removal; verification touches all 6

CCNA 200-301: Domain 3.0 (IP Connectivity)

Purpose: the actual payoff of this entire phase. Remove the Phase 1 stopgap static default routes — OSPF now provides specific routes to every internal subnet, so the default route is no longer needed for internal reachability.

Commands

```
! PH-MNL-ROAS:
no ip route 0.0.0.0 0.0.0.0 10.10.254.5

! TH-BKK-ROAS:
no ip route 0.0.0.0 0.0.0.0 10.10.254.9
```

Word / term	Meaning
no ip route ...	Removes a previously configured static route. The exact same destination/next-hop pair must be specified to remove it — IOS doesn't have a "delete the last route I added" shortcut.

DO THIS IN THE RIGHT ORDER — VERIFY BEFORE REMOVING

Confirm OSPF has actually learned real replacement routes *before* deleting the static ones, otherwise reachability is briefly cut with nothing to replace it:

```
show ip route ospf
```

Only remove the static route once this shows real 0 (OSPF-learned) entries for the subnets that route was covering.

EXPECTED RESULT — THE REAL PROOF

```
ping 10.10.120.1    (from PH-MNL-ACC, to Bangkok's gateway)
ping 10.10.110.1    (from TH-BKK-ACC, to Manila's gateway)
```

Both should now succeed — cross-site reachability that flatly did not work at the end of Phase 1. This is the single most important verification in the whole project so far: three separate countries' networks becoming one, entirely through dynamic routing, with no static routes holding it together.

A RELATED LIMITATION WORTH UNDERSTANDING, NOT A BUG

Pinging `MY-KL-HQ-CORE`'s own LAN gateways (e.g. `10.10.10.1`) cross-site fails — not an OSPF problem, but because those VLAN SVIs show `Status: up`, `Protocol: down` in this lab (no live end device, WLC, or distribution switch is actually connected to those VLANs, a Phase 1 topology limitation). With no active port in that VLAN, OSPF has nothing to advertise for that subnet at all — there is no route to it anywhere in the network, regardless of how correct the OSPF configuration is. Connecting a real test PC to one of `MY-KL-HQ-CORE`'s access ports would resolve this instantly by bringing that VLAN's protocol state up.

8. Objective 6 — IPv6 Addressing on Interfaces

MY-KL-HQ-CORE, SG-EDGE-GW, PH-MNL-ROAS, TH-BKK-ROAS

CCNA 200-301: Domain 1.0 (Network Fundamentals — IPv6 addressing)

Purpose: apply the addressing plan from Objective 1 to real interfaces, completing the dual-stack (IPv4 + IPv6 simultaneously) design across every routing device.

Commands, word by word

```
ipv6 unicast-routing
!
interface Vlan10
  ipv6 address 2001:DB8:1:10::/64 eui-64
!
interface GigabitEthernet1/0/24
  ipv6 address 2001:DB8:FF::0/127
```

Word / term	Meaning
ipv6 unicast-routing	Required globally before this device will forward IPv6 traffic between interfaces at all — off by default even on a device with IPv6 addresses already assigned, the same conceptual gap as <code>ip routing</code> being required for IPv4 on a multilayer switch in Phase 1.
eui-64	Short for modified EUI-64 (Extended Unique Identifier, 64-bit) — an algorithm that derives the last 64 bits of an IPv6 address (the interface identifier) automatically from the interface's 48-bit MAC address, by splitting it in half and inserting the fixed value <code>FFFE</code> in the middle. You supply the /64 network prefix; the router computes the host portion itself.
Explicit addressing on WAN links (no eui-64)	For a fixed 2-address point-to-point /127 link, manually assigning the two addresses (<code>::0</code> and <code>::1</code> , etc.) is simpler than deriving them from a MAC address, and matches real RFC 6164 practice.

EXPECTED RESULT

```
show ipv6 interface brief
```

Every addressed interface should show both the global `2001:db8:...` address just typed and an automatically-generated link-local `fe80:...` address.

9. Platform Lessons Learned

Observation	Resolution
<code>auto-cost reference-bandwidth 10000</code> found missing on <code>MY-KL-HQ-CORE</code> during the post-typing audit	Genuine typing omission (not a platform limitation) — added and reverified via <code>show run section router ospf</code>
<code>log-adjacency-changes</code> appearing in <code>show run</code> under the OSPF process, despite never being typed	An automatic default that's been enabled on OSPF since fairly old IOS releases and gets explicitly shown as informational state — not a configuration gap, same pattern as <code>spanning-tree extend system-id</code> from Phase 1
A configured IPv6 address like <code>2001:DB8:FF::0/127</code> displaying in <code>show run</code> as <code>2001:DB8:FF::/127</code>	Identical address, not a mismatch — <code>::</code> already absorbs a trailing all-zero group, so an explicit trailing <code>::0</code> is redundant and IOS normalizes it away when displaying

10. Real Bugs Found During Verification

Device	Bug	Severity	Fix
MY-KL-HQ-CORE	Missing auto-cost reference-bandwidth 10000 under the OSPF process	Moderate — path-cost math inconsistent with the other 3 routers until fixed	Added the missing line, reverified
MY-KL-HQ-CORE (live cabling)	The physical WAN port wiring to Manila/Bangkok ended up swapped relative to the original design assumption (Manila landed on Gi1/0/23 , Bangkok on Gi1/0/22 , opposite of the first draft)	Cosmetic once corrected — descriptions/IPs were realigned to match actual cabling rather than forcing a re-cable	Swapped the IP/description assignment between the two interfaces to match reality, propagated the correction through the topology docs and both ROAS routers' WAN-link descriptions

11. Appendix — Verification Command Quick Reference

What to check	Command
OSPF neighbor state	<code>show ip ospf neighbor</code>
OSPF process settings, passive interfaces, Router ID	<code>show ip protocols</code>
OSPF-learned routes	<code>show ip route ospf</code>
Per-interface OSPF cost, priority, DR/BDR role	<code>show ip ospf interface <if></code>
IPv6 addressing	<code>show ipv6 interface brief</code>
Cross-site reachability proof	<code>ping <remote-site-gateway></code>

AREHI-SECOPS — Phase 2 Full-Depth Configuration Guide · Generated as a certification-study companion to the `01-regional-on-premises-network/` configs and topology documentation in this repository.